

Optimal Allocation of CHP-Based Distributed Generation on Urban Energy Distribution Networks

Xianjun Zhang, *Student Member, IEEE*, George G. Karady, *Life Fellow, IEEE*, and Samuel T. Ariaratnam

Abstract—The continuing penetration of combined heat and power (CHP)-based distributed generation (DG) has made urban electricity, water, and natural gas distribution networks increasingly interconnected. This paper analyzes the optimal network capacity and distribution of the CHP-based DG based on urban energy distribution networks by introducing an integrated system dispatch model. The electricity, water, and gas network models were designed and developed individually. The CHP-based DG model was developed to couple these energy distribution systems. The results indicate the optimal allocation of CHP-based DG by analyzing multiple factors and mutual impacts on operational performances of the CHP-based DG units and energy distribution networks. The designed typical gas system is capable of supplying sufficient natural gas for DG normal operations, whereas the present water system cannot support the complete recovery of the exhaust heat from large CHP-based DG penetration.

Index Terms—Combined heat and power (CHP), distributed generation (DG), load flow analysis, mathematical programming, water and gas distribution networks.

NOMENCLATURE

Electrical System

<i>Set:</i>		
C	Set of shunt capacitors (indexed by c).	
G	Set of new generators (indexed by g).	
L	Set of electrical lines (indexed by l).	
N	Set of electrical nodes (indexed by n).	
X	Set of external sources (indexed by x).	
<i>Parameters:</i>		
$C_{c,n}^{(+,-)}$	(Max, Min) VAr capacity of c at n .	
$D_n^{(P,Q)}$	(Real, Reactive) power demand at n .	
F_l^+	Max MVA capacity of l .	
$(G, B)_L$	(Conductance, Susceptance) of l .	
$N_{(l,b)}^{(A,B)}$	(From, To) node of l and b .	
$(P, Q)_{x,n}^{X,(+,-)}$	(Max, Min), (Real, Reactive) power of x at n .	
$V_n^{(+,-)}$	(Max, Min) voltage at n .	
$\phi_{(g,x)}$	Power angles of g and x .	
$\beta_{(c,g,p,v,w,x)}^N$	(c, g, p, v, w, x) location corresponding to n .	
<i>Variables:</i>		
$f_l^{(A,B),(P,Q)}$	(Real, Reactive) power injection onto l at (From, To) node.	
$p_{g,n}^{(E,T)}$	(Electrical, Thermal) output of g at n .	
$(p, q)_{x,n}^X$	(Real, Reactive) power output of x at n .	
$s_{c,n}^C$	VAr injection by c at n .	
$(v, \delta)_n$	(Voltage Level, Voltage Phase) at n .	
<i>Gas System</i>		
<i>Parameters:</i>		
$(A, B, C)^{Co}$	Coefficient of DG's power/fuel equation.	
$(C, \rho)^W$	Water heat capacity and density.	
$F_{g,n}$	Natural gas consumption by g at n .	
$P^{(E,T)}$	(Electrical, Thermal) capacity of DG unit.	
$P_{g,n}^{(E,T),(+,-)}$	(Max, Min), (Electrical, Thermal) power output of g at n .	
T^A	Ambient temperature.	
$T_{g,n}^{(SD,RT),(+,-)}$	(Max, Min) (Sending, Returning) water temperature of g at n .	
$W_{g,n}^{F(+,-)}$	(Max, Min) water flow of g at n .	
η^{HE}	Heat exchanger efficiency.	
<i>Variables:</i>		
$cJ_{g,n}^{(WT,WF)}$	Correction factor of (Water Temperature, Water Flow) of g at n .	
$p_{g,n}^{(FE,FT)}$	Final (Electrical, Thermal) output of g at n .	
$t_{g,n}^{(SD,RT)}$	(Sending, Returning) water temperature of g at n .	
$w_{g,n}^D$	Water demand by DG g at n .	
<i>Water System</i>		
<i>Set:</i>		
B	Set of water network links (indexed by b).	
P	Set of pumps (indexed by p).	
V	Set of valves (indexed by v).	
W	Set of water sources (indexed by w).	
<i>Parameters:</i>		
B_b^R	Link b resistance coefficient.	
C^{HW}	Hazen–Williams roughness coefficient.	
$(D, L)_b$	Link b (Diameter, Length).	
K^V	Energy loss coefficient of v .	
P_p^{Co}	Pump p operation coefficient	
W_n^D	Existing water demand at n .	
$W_n^{H(+,-)}$	(Max, Min) water head pressure at n .	
$\beta_{(p,v)}^B$	(p, v) location corresponding to b .	

Manuscript received March 11, 2013; revised July 12, 2013; accepted August 12, 2013. Date of publication October 10, 2013; date of current version December 12, 2013. This work was supported by the U.S. National Science Foundation (NSF), the Emerging Frontiers in Research and Innovation (EFRI) under Grant 0836046.

X. Zhang and G. G. Karady are with the School of Electrical, Computer, and Energy Engineering, Arizona State University, Tempe, AZ 85287 USA (e-mail: xianjun.zhang@asu.edu; karady@asu.edu).

S. T. Ariaratnam is with the School of Sustainable Engineering and the Built Environment, Arizona State University, Tempe, AZ 85287 USA (email: ariaratnam@asu.edu).

Digital Object Identifier 10.1109/TSTE.2013.2278693

Variables:

$r_{w,n}^W$	Water supplied by w at n .
w_b^F	Water flow in link b .
w_n^H	Water head at n .

I. INTRODUCTION

A WORLDWIDE energy crisis and growing environmental concerns are driving the development and penetration of distributed generation (DG) into distribution systems. DG technologies provide solutions for requesting efficient, environmentally friendly, economic, and reliable power sources [1]–[3]. Demand reduction through improved energy efficiency is an essential metric in engineering resilient and sustainable urban energy infrastructures for continual urban growth. One effective and efficient means of promoting energy efficiency is the employment of combined heat and power (CHP) based on DG technologies.

Many gas-fired DG units, such as microturbines, fuel cells, and internal combustion engines, can provide thermal energy by recovering exhaust heat during the process of power generation. This thermal energy can boost total energy efficiency up to 85% by providing hot water, air conditioning, or district heat supply [4], [5]. In the United States, the average energy efficiency of typical coal and natural gas power plants is only in the range of 20%–38% and 30%–50%, respectively [6], [7]. Although gas-fired DG units may use natural gas as the fuel, other crude fuels such as biomass can also make for superior substitutes. The CHP-based DG is intended to supply multiple consumers instead of the individual consumer with one unit for each, which is favored and verified by operating cost optimizations as in [8].

The urban electricity, water, and natural gas distribution networks are becoming more interconnected with the increased coupling of CHP-based DG units. These energy distribution infrastructures explicitly affect the accommodated CHP-based DG [3], [9]. In this context, the optimal allocation of CHP-based DG mainly investigates the problems of sizing, siting, and mutual operational impacts. A wide variety of mathematical optimization-based studies investigate the DG siting and sizing problems. In [10] and [11], an optimal power-flow (ACOPF)-based approaches are used to analyze the accommodated DG network capacity, in which the DG connection points are fixed and the maximum DG power injections are evaluated. A multi-objective evolutionary algorithm is proposed to determine the DG sizing and siting in [12]. Optimization models with DG integration for radial distribution system planning and expansion are presented in [13] and [14]. Reference [15] introduces a new optimization model to obtain the optimal DG sizing and siting based on a cost-benefit analysis approach.

A number of studies have investigated the interactions between DG, electricity, heat, and natural gas networks. In [16] and [17], an approach based on the combined power flows of multiple energy infrastructures, e.g., electricity, gas, and district heating, is presented and the concept of the energy hub is introduced to deal with integrated energy infrastructures. A three-node combined system of electricity and gas is used to examine the developed optimization model. Most prior

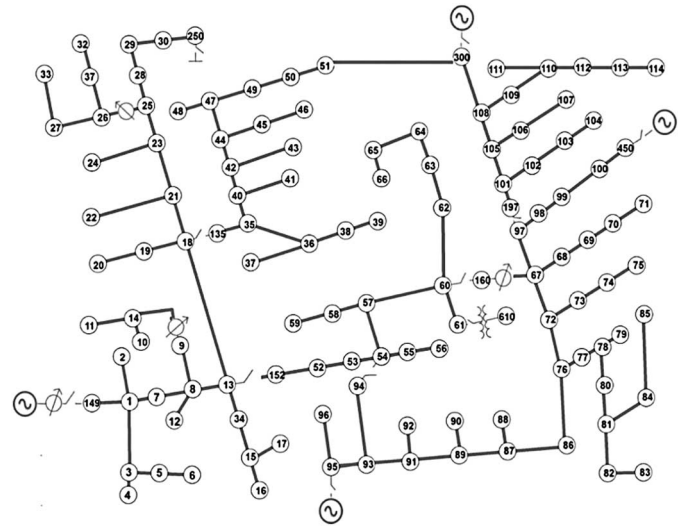


Fig. 1. Modified electricity network.

investigations have focused on the issues of interactions between electricity and natural gas networks at a transmission level rather than at the distribution level. In [18] and [19], power-flow-based analyses and models are developed for the integrated natural gas and electricity networks. References [3] and [9] present an analysis of the impacts of gas and water distribution networks on DG siting and sizing. An integrated gas electricity OPF formulation and an integrated heat electricity OPF formulation are described in [20] and [21], respectively, with the DG connection points fixed.

In the current literature, however, relatively large natural gas and water distribution networks, corresponding to an electricity distribution network, have not yet been modeled with the coupling of the CHP-based DG in integrated energy optimization models. An analysis of gas and water at the distribution level is a much less mature subject than that in centralized power systems at the transmission level, especially in the context of CHP-based DG penetration. Therefore, to address the challenges of 1) identifying the optimal sizing and siting of CHP-based DG based on urban energy networks and 2) quantifying the mutual impacts on operational performance, a clear need emerges for the development of a model architecture that integrates these energy networks with the CHP-based DG. This is especially valuable as integrated decisions can be made to provide a valuable guidance for urban energy planning activities and thus directly benefit investments made by distribution network operators.

II. ELECTRICAL SYSTEM MODEL DEVELOPMENT

For this study, the typical radial distribution feeder, IEEE 123 node test feeder [22], was selected and modified into a looped system by adding external ac power sources (ExGens) and by connecting node 51 and node 300, as shown in Fig. 1.

The modified system is not a completely three-phase system, thus the ACOPF formulation was applied to each phase to accurately evaluate the results.

1) *Capacity constraints of CHP-based DG units:* The CHP-based DG units are assumed to be placed at each node. For electrical and thermal capacities of CHP-based DG units, the

minima are defaulted as zero and the maxima should be prorated with the total capacity changing of nodal DG units:

$$P_{g,n}^{(E,T),-} \leq p_{g,n}^{(E,T)} \leq P_{g,n}^{(E,T),+} \quad \forall g \in G \quad \forall n \in N. \quad (1)$$

2) Capacity constraints of ExGens:

$$(P, Q)_{x,n}^{X,-} \leq (p, q)_{x,n}^X \leq (P, Q)_{x,n}^{X,+} \quad \phi_x = 0 \\ \forall x \in X \quad \forall n \in N. \quad (2)$$

3) Nodal voltage level constraint:

$$V_n^- \leq v_n \leq V_n^+ \quad \forall n \in N. \quad (3)$$

4) Capacity constraint of shunt capacitors:

$$C_{c,n}^- \leq s_{c,n}^C \leq C_{c,n}^+ \quad \forall c \in C \quad \forall n \in N. \quad (4)$$

5) Kirchhoff voltage law constraints [10] $\forall l \in L$:

$$f_l^{A,P} = G_l (v_{N_l^A})^2 - v_{N_l^A} v_{N_l^B} \\ \left[G_l \cos(\delta_{N_l^A} - \delta_{N_l^B}) + B_l \sin(\delta_{N_l^A} - \delta_{N_l^B}) \right] \quad (5)$$

$$f_l^{A,Q} = B_l (v_{N_l^A})^2 - v_{N_l^A} v_{N_l^B} \\ \left[G_l \sin(\delta_{N_l^A} - \delta_{N_l^B}) + B_l \cos(\delta_{N_l^A} - \delta_{N_l^B}) \right]. \quad (6)$$

Apply constraints (5) and (6) to the other end of a line.

6) Nodal real and reactive power balance constraints:

$$\sum_{\forall x \in X | \beta_x^N = n} p_{x,n}^X + \sum_{\forall g \in G | \beta_g^N = n} p_{g,n}^E \\ = D_n^P + \sum_{\forall l \in L | N_l^A = n} f_{l,n}^P + \sum_{\forall l \in L | N_l^B = n} f_{l,n}^P \quad (7)$$

$$\sum_{\forall x \in X | \beta_x^N = n} q_{x,n}^X + \sum_{\forall g \in G | \beta_g^N = n} p_{g,n}^E \tan(\phi_g) + \sum_{\forall c \in C | \beta_c^N = n} s_{c,n}^C \\ = D_n^Q + \sum_{\forall l \in L | N_l^A = n} f_{l,n}^Q + \sum_{\forall l \in L | N_l^B = n} f_{l,n}^Q \\ \forall n \in N. \quad (8)$$

7) Thermal constraints of lines:

$$\left(f_l^{(A,B),P} \right)^2 + \left(f_l^{(A,B),Q} \right)^2 \leq (F_l^+)^2 \quad \forall l \in L. \quad (9)$$

III. NATURAL GAS SYSTEM MODEL DEVELOPMENT

According to the electrical network, a representative natural gas distribution network was designed and modeled based on commercial gas simulation software [23], as shown in Fig. 2. It was assumed that for the spot load where electricity was supplied, gas was also provided. Each node in the gas network corresponds to the same node in the electrical system.

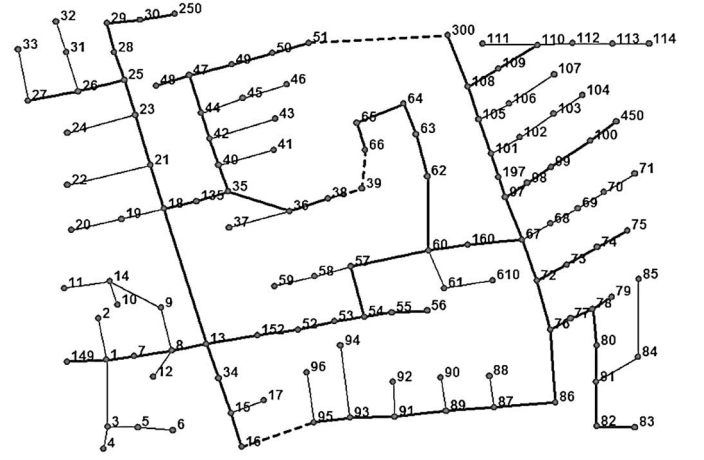


Fig. 2. Designed natural gas distribution network.

In Fig. 2, dashed lines are added to connect gas mains to improve the gas supply reliability by changing the original radial system into a looped system. Service lines less than 1 inch in diameter are not modeled in this network. Valves are not modeled in this network as they are only used to switch the gas supply ON or OFF in a certain area, most commonly during emergencies. The regulators are used in the distribution system to reduce the output gas pressure and thus not modeled. Node 250 is designated as the supply node (city gate) of natural gas, with the pressure fixed at 60 lb per square inch gauge (psig). The downstream gas pressure after this node in the distribution is below 60 psig. The elevation in the gas network was assumed the same. The gas pressure in the residential area could be 0.25 psig or lower.

The base nodal gas consumption of the clustered load was calculated by multiplying the number of nodal households by the hourly gas consumption of an individual household. The 2005 per capita residential gas consumption in Arizona was 6.1 million Btu [24]. An assumption of 3.2 people per household was made, along with an average usage time of 3 h/day. The average household gas consumption was then calculated as 19.52 thousand cubic feet (mcf) annually and 17.83 cubic feet/h (0.504 m³/h). Therefore, the total base gas load was 888.33 m³/h for the designed gas distribution network. For the designed gas piping system, to calculate the gas flow and pressure drop in pipelines, the Institute of Gas Technology (IGT)-Improved Equation [25] was selected and used in this application of medium and high pressure distribution (1–500 psig), with the operation of 2 inches and larger diameter pipes.

IV. WATER SYSTEM MODEL DEVELOPMENT

Corresponding to the electrical system, a generalized water distribution network was designed, and its mathematical model is developed in this section.

A. Water Network Design

Each node in the water network was assumed to represent a cluster of houses. The nodal water demand was estimated based

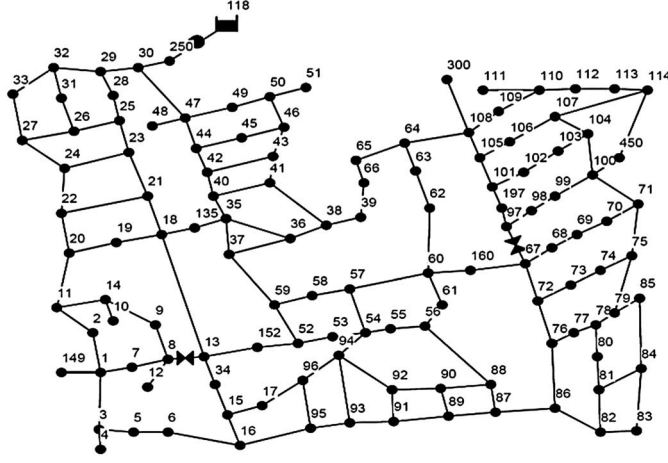


Fig. 3. Designed water distribution network.

on the number of nodal households and the nodal electrical load. This representative water distribution network, as shown in Fig. 3, was designed based on the cost minimization to choose appropriate pipe diameters, for which the details can be found in [9]. A reservoir is located at node 118 to provide all the nodal water demands. A pump located between node 118 and node 250 increases the water head. Two pressure reducing valves (PRV) are also incorporated in this network. The elevation of all nodes was assumed the same except the reservoir node to neglect the gravity impact on water flow.

B. Water Network Model Development

To effectively integrate the water system and the electrical system, a mathematical water network model was also developed based on the designed water system.

1) *Nodal water head constraint*: For all nodes during peak hours, the minimum nodal water head may be changed for case studies

$$W_n^{H-} \leq w_n^H \leq W_n^{H+} \quad \forall n \in N. \quad (10)$$

2) *Pump equation* $\forall p \in P \forall b \in \text{pumplink}$: By using EPANET, the pump equation was generated based on one point (1140 m³/h, 45 m of water head) acquired from the pump curve to represent the relationship between water head gain and water flow in the pump link

$$w_{p|\beta_p^N=N_b^A}^H - W_{p|\beta_p^N=N_b^B}^H = W_{\text{MinDsgn}}^H - P_p^{Co} \times (w_{\beta_p^B}^F)^2. \quad (11)$$

3) *Water head drop* $\forall b \in B$ except the pump link: The water head drops when flowing through pipelines [26]

$$w_{N_b^A}^H - w_{N_b^B}^H = B_b^R \times (w_b^F)^{1.85} \quad (12)$$

$$B_b^R = 1.439 \times (C^{HW})^{-1.85} \times (D_b)^{-4.87} \times L_b. \quad (13)$$

4) *Pressure reducing valves*: The PRVs [26] restrict the outflow from exceeding 60 m

$$w_{v|\beta_v^N=N_b^A}^H - w_{v|\beta_v^N=N_b^B}^H = 6381.94 \times \frac{K^V \times w^F (\beta_v^B)^2}{(D_{\beta_v^B})^4} \quad \forall l = L. \quad (14)$$

5) *Nodal balance equation*: The water flow into a node equals the nodal water demand plus the water flow out of the node and the extra water demand resulting from the employment of CHP-based DG

$$\begin{aligned} \sum_{\forall w \in W|\beta_w^N=n} r_{w,n}^W - \sum_{\forall b \in B|\beta_b^N=n} w_b^F + \sum_{\forall b \in B|\beta_b^N=n} w_b^F \\ = \sum_{\forall g \in G|\beta_g^N=n} w_{g,n}^D + W_n^D \quad \forall n \in N. \end{aligned} \quad (15)$$

V. THE CHP-BASED DG MODEL DEVELOPMENT

A. CHP-Based Microturbines

The microturbine C65 [27] was selected as a reference for the CHP-based DG unit. The capacities of these series of products can be modularized, which show similar electrical and thermal characteristics as the selected reference unit.

The exhaust heat from microturbines travels through heat exchangers, where pressurized water flows and carries away exhaust heat. This water flow operates as a looped circulating process and a certain amount of water is released from the water network periodically or during the peak thermal demand.

B. Mathematical Formulation

The developed mathematical formulation takes into account the influence of the microturbine's fuel consumption, water flow rate, water temperatures, and ambient temperatures, as shown in the following.

1) *Power/gas conversion equation* $\forall g \in G \forall n \in N$: One quadratic equation was derived from the dataset of the DG unit's fuel consumption rate and corresponding power output [28], which is found best fit for the DG fuel consumption trend with $R^2 = 0.9996$

$$F_{g,n} = A^{Co} \times (p_{g,n}^E)^2 + B^{Co} \times p_{g,n}^E + C^{Co}. \quad (16)$$

2) *DG water consumption constraints*: The actual maximum/minimum nodal DG water consumption should be prorated with the total capacity changing of nodal DG units

$$W_{g,n}^{F-} \leq w_{g,n}^D \leq W_{g,n}^{F+} \quad \forall g \in G \forall n \in N. \quad (17)$$

3) *Water temperature constraints*: For selected CHP-based DG units, the sending and returning water temperatures can be found in [28]

$$T_{g,n}^{(SD,RT),-} \leq t_{g,n}^{(SD,RT)} \leq T_{g,n}^{(SD,RT),+} \quad \forall g \in G \forall n \in N. \quad (18)$$

4) *Heat transfer equation* $\forall g \in G \forall n \in N$: The transferred heat power from the DG exhaust heat also depends on water mass flow and temperature difference [21]

$$3600 \times \eta_{HE} \times p_{g,n}^T = C^W \times \rho^W \times w_{g,n}^D \times (t_{g,n}^{SD} - t_{g,n}^{RT}). \quad (19)$$

5) *Ambient temperature constraints* $\forall g \in G \forall n \in N$: The relationships between ambient temperature and DG output are described by (20) and (21), respectively, which are extracted from the DG output curves with the changing of T^A [28]

$$T^A > 70^\circ F \begin{cases} P_{g,n}^T \leq \frac{P_{g,n}^E}{P^E} [P^T - 0.1905(T^A - 70)] \\ P_{g,n}^{FE} = P_{g,n}^E [1 - 0.00577(T^A - 70)] \end{cases} \quad (20)$$

$$T^A \leq 70^\circ F \begin{cases} P_{g,n}^T \leq \frac{P_{g,n}^E}{P^E} [P^T - 0.4308(70 - T^A)] \\ P_{g,n}^{FE} = P_{g,n}^E \end{cases}. \quad (21)$$

6) *Correction factors constraints* $\forall g \in G \forall n \in N$: The recovered DG thermal output is also affected by the inlet water temperature and the water flow rate. The correction factors are thus generated for the recovered DG thermal output [28]

$$cf_{g,n}^{WT} = 1 - 0.0023 \times (t_{g,n}^{RT} - 140) \quad (22)$$

$$cf_{g,n}^{WF} = 0.83 + 0.02464 \times w_{g,n}^D - 0.0001496 \times (w_{g,n}^D)^2. \quad (23)$$

7) *Final DG recovered thermal output*:

$$p_{g,n}^{FT} = p_{g,n}^T \times \eta^{HE} \times cf_{g,n}^{WT} \times cf_{g,n}^{WF} \quad \forall g \in G \forall n \in N. \quad (24)$$

VI. PROBLEM SOLVING, CASE STUDIES, AND DISCUSSION

A. Objective

The overall objective of the integrated energy distribution system is to maximize the DG electrical output and the recovered thermal output. The electrical output is constrained by the electrical and gas systems, whereas the recovered thermal output is constrained by the electrical system, water system, and the CHP-based DG system

$$\max \sum_{g \in G} \sum_{n \in N} (p_{g,n}^{FE} + p_{g,n}^{FT}). \quad (25)$$

B. Implementation

The whole mathematical model was coded and implemented as a nonlinear mathematical programming problem using AIMMS [29]. The CONOPT 3.14-V generalized reduced gradient solver was selected. The nonlinear presolver and multistart procedure [29] were also employed to reduce problem size, search space, and increase the chances of finding the global optimum by comparing all found feasible solutions.

The problem-solving flowchart of the integrated system is shown in Fig. 4. The ACOPF-based electrical system model is optimized first to find the maximum nodal DG electrical output and thermal output, which are then constrained and updated by the developed gas system. The updated nodal DG electrical

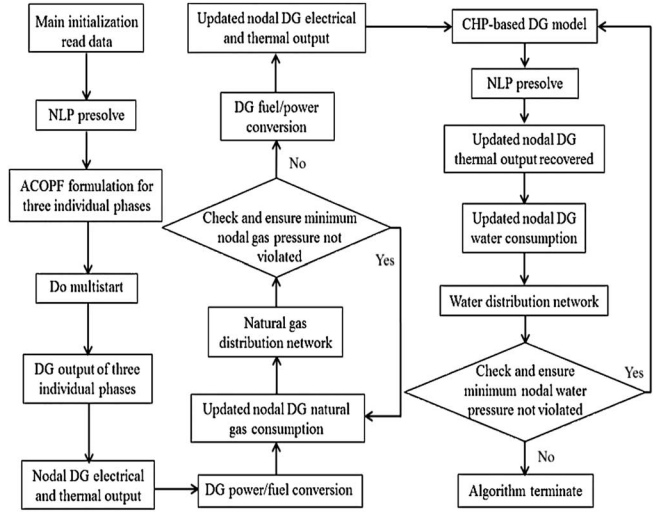


Fig. 4. Integrated system solving flowchart.

output and thermal output are then used as input parameters for the CHP-based DG model and water network model, where they are further constrained and finally determined. If the minimum nodal water pressure is violated, then less water will be available to facilitate the thermal heat recovery, resulting in less recovered DG thermal output. This iteration continues until no nodal water pressure violation occurs.

C. Case Studies

Multiple factors affect the CHP-based DG units' output, such as capacity bounds, power factor (PF), gas supply, ambient temperature, and nodal water pressure.

1) *Capacity Bounds*: By changing the capacity of DG units to 65, 130, 325, 650, and 1 MW, five cases from Case 1 to Case 5 were studied to investigate the influence of the capacity change on DG performance and distribution. A unity PF was assumed for each case. For this analysis, only the electrical system model was optimized, whereas the gas-, water-, and CHP-based DG system models were not integrated.

The electrical output summary for this study is shown in Table I. The total DG electrical output rises as the unit capacity increases. The electrical system load is 3.49 MW and thus the grids absorb the extra DG real power output. The base natural gas load for the designed gas network is 31.089 mcf/h. The total DG gas consumption is found at least 5.91 times more than the base gas usage for Case 1.

The DG penetration level refers to the ratio between the number of DG nodes and the total number of nodes. From Table I, it is shown that not every node is appropriate to place DG units and the penetration level reduces with the increased DG capacity of units. The reason for this is that with increasing capacity of DG units, a DG node continues to raise power output until a violation occurs at that node. The penetration level should not be too high or too low. If the penetration level is too high, such as in Case 1, then the potential fault level and stability issues in the distribution system grow significantly. If the penetration level is too low, such as in Case 5, the economic benefits may not be sufficiently materialized, e.g., the paving of new thermal pipelines is costly when the load center is located far away from the thermal source.

TABLE I
ELECTRICAL OUTPUT SUMMARY

Case studies	Case 1	Case 2	Case 3	Case 4	Case 5
DG power output (MW)	16.84	23.09	25.22	28.49	32.02
DG penetration level (%)	100	89.3	75.4	62.3	59
Grids power output (MW)	-13.35	-19.6	-21.73	-25	-28.53
DG power supply ratio	4.82	6.62	7.23	8.16	9.17
DG gas usage (mcf/h)	183.82	245.70	265.36	296.73	331.98
DG gas use ratio	5.91	7.90	8.54	9.54	10.68

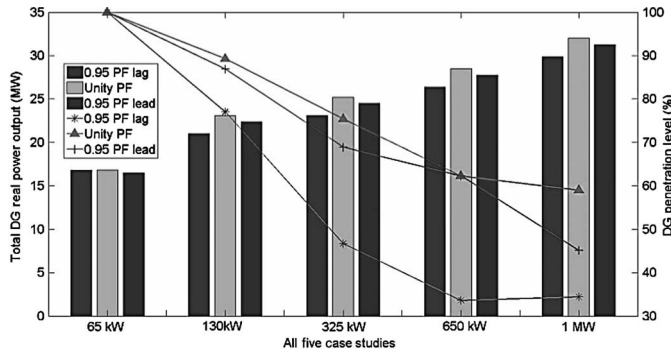


Fig. 5. DG power output and penetration level with changing PFs in all cases.

2) *Power Factor*: Three different DG PFs, 0.95 lag, unity, and 0.95 lead, were analyzed for all five cases and only the electrical system was investigated. As shown in Fig. 5, the overall trend of total DG power output is growing with the unit capacity increasing for all cases. For each case, DG outputs the most power at the unity PF while outputting the least power at the 0.95-PF lag. For Case 3, the maximum DG power output is 25.22 MW. Fig. 5 also shows that the DG penetration level reduces continuously at the unity PF. However, for the 0.95-PF lead, the decreasing trend of the DG penetration level slows down, especially when the unit capacity is beyond 325 kW. The same situation happens to the DG penetration level at the 0.95-PF lag, which decreases the most (down to 33%) and remains nearly level after 650 kW.

For PFs of 0.95 lag and unity, the reactive power is supplied by the electrical system and all the shunt capacitors work at full capacities. When the PF is 0.95 lead, the electrical system absorbs the extra DG reactive power output. The PF of maximum DG units can be specified according to the customer requirements.

3) *Gas Supply*: All five cases in the electrical system were optimized and used in the gas system, and Case 3 was analyzed with three different gas main sizes. The water system model was not integrated. The nodal gas pressure curves are shown in Fig. 6. For all cases, the resultant nodal gas pressure is acceptable for the normal operation of DG units as this pressure can be as low as 0.25 psig with the integration of a gas compressor or fuel booster. It was also found that the higher the spike of the curve, the higher the nodal gas pressure and correspondingly, less nodal gas was consumed, such as node ID# around 30 and 120. Conversely, the pressure sag indicates nodes where more gas was consumed compared to other nodes. From Case 1 to Case 5, since the DG power output increases, the total gas consumption also increases and thus the nodal gas pressure drops accordingly.

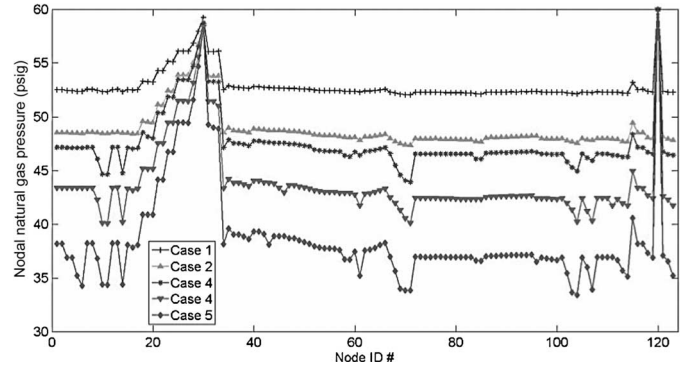


Fig. 6. Nodal natural gas pressure with the DG distribution.

TABLE II
GAS SYSTEM PERFORMANCES FOR DIFFERENT GAS MAINS

Gas main size/type	4P				
Scaled total gas load (mcf/h)	184.76	187.65	190.53	193.42	196.31
Nodes with positive gas pressure (%)	100	95.9	50.4	29.3	22.0
Gas load supplied (mcf/h)	184.76	175.55	80.63	46.68	34.88
DG gas usage (mcf/h)	153.67	144.47	49.54	15.59	3.79
DG power output (MW)	15.17	14.26	4.88	1.53	0.36
Gas main size/type	2P				
Scaled total gas load (mcf/h)	32.91	33.20	33.49	33.78	34.06
Nodes with positive gas pressure (%)	100	67.5	48.8	44.7	27.6
Gas load supplied (mcf/h)	32.91	20.32	12.52	10.52	8.16
DG gas usage (mcf/h)	1.82	0	0	0	0
DG power output (MW)	0.17	0	0	0	0

For Case 3, the total gas load of 296.45 mcf/h was scaled down to different values. Also, the previous gas main diameter of PE 5.314 (6P) inches was changed to PE 3.621 (4P) inches and PE 1.885 (2P) inches, respectively. The results are shown in Table II to study the influence of gas main diameter on gas supply and DG power output. For both 4P and 2P cases, the ratio of nodes with positive gas pressure reduces up to 78% with a small increase (up to 6.25%) in the scaled total gas load. Nodes with negative gas pressure cannot supply gas such that the total supplied gas load decreases with an increase in negative gas pressure nodes. In either case, the base gas load has the priority of being first supplied, which further reduces the remaining gas for the DG usage. For the 4P case, the maximum DG power output is 15.17 MW compared to 25.22 MW in the 6P case, whereas for the 2P case, the maximum DG power output is only 0.17 MW, about 0.67% of that in the 6P case. The whole gas network in this case does not work and besides cannot support the DG penetration. Therefore, a larger diameter of the gas main increases the line pack capability and compensates for fluctuations in natural gas.

4) *Ambient Temperature*: For this case study, all four system models were integrated and optimized. Case 3 (0.325 MW) and a unity PF were selected. The nodal water head was set as 40 m. A range of different ambient temperatures T^A (0 °F–122 °F) are selected as references to study their influence on DG performances.

The total DG electrical output, recovered thermal output, and total power output curves are shown in Fig. 7. It is found that ambient temperatures significantly affect the total DG power output. When T^A is below 70 °F, the electrical output remains

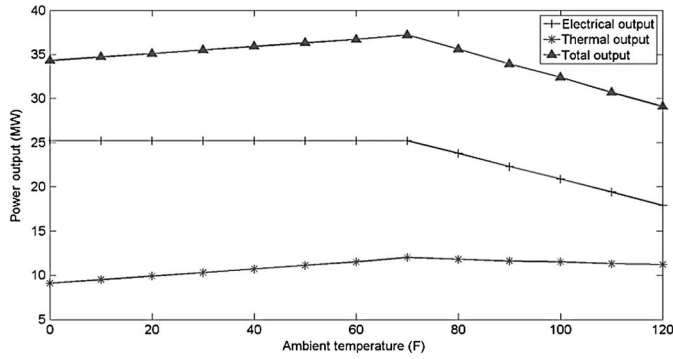


Fig. 7. Total DG power output with changing ambient temperatures.

unchanged while the recovered thermal output decreases with the reduction of T^A . When T^A is above 70 °F, the electrical output and the recovered thermal output decrease to 5.78% and 1.39%, respectively, for every 10 °F increase. The total power output reaches the maximum 37.24 MW at T^A of 70 °F.

5) *Nodal Water Pressure*: The nodal water pressure is the main factor constraining water consumption and thus DG recovered thermal output. For this case study, all four system models were integrated together and optimized. Case 3 (0.325 MW) and a unity PF were selected. T^A was set to 70 °F. A water head above 35.2 m is acceptable in this study and several other water heads, 36, 38, 40, and 42 m, are also selected for comparison purposes.

For all nodal water head limits, the electrical output remains unchanged as 25.22 MW, whereas the recovered thermal output decreases significantly from 23.70 to 4.21 MW, with an increase in the nodal water head limit from 36 to 42 m. The maximal total DG output is thus reduced from 48.91 to 29.43 MW with a water head limit ranging from 36 to 42 m.

The nodal water head curves with different limit requirements are shown in Fig. 8. The nodal water heads with node ID# between 20 and 30 are higher compared to the other nodes except for nodes clustered around 120, which are located close to the pump. For node ID# between 70 and 110, they correspond to the nodes with lower water pressures as they are located at the far right side of the pump. The water head decreases automatically with the increasing distance from the pump. For a single node, it is also found that the more the water extracted, the lower the nodal water head. The application and penetration of the CHP-based DG increase with a reduction in the nodal water head limit. The water system is considered to be functional for additional DG water demand during peak hours if an acceptable pressure drop of 10 psig is allowed.

The recovered thermal output is summarized in Table III. Typically, 13% of the electricity [3] is used to heat the household water, which is 0.4537 MW for the electrical system in this study. It is known that the total base hot water load is 342.97 m³/h, and the total base water load is 1143.24 m³/h. The hot water usage ratio compared to the total base water load is calculated and the results indicate that the hot water supply is only partially met from 10.2% to 54.5%. Therefore, the money saved is prorated as shown in Table III. If the DG units supply all the required hot water, then the money saved is calculated from \$0.07 to \$2.22 per household per day. Savings accrue by directly using the hot water supplied by the DG units, which would otherwise have been bought from grids.

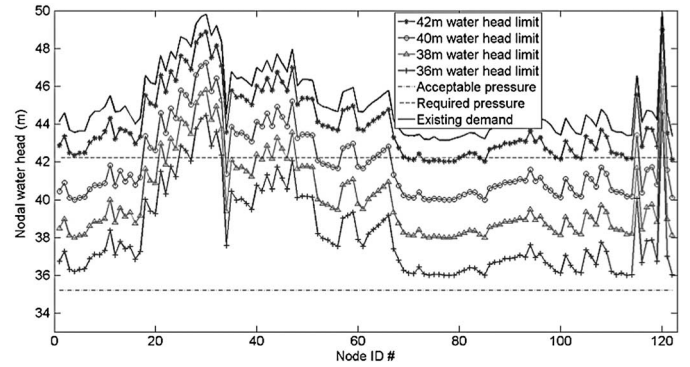


Fig. 8. Nodal water head with different water head limits.

TABLE III
RECOVERED THERMAL OUTPUT SUMMARY

Nodal water head limit (m)	36	38	40	42
Total DG hot water supply (m ³ /h)	187.01	142.78	93.20	35.14
DG hot water usage ratio (%)	16.4	12.5	8.2	3.1
Hot water serving ratio (%)	54.5	41.6	27.2	10.2
Daily money saved per household (\$)	2.22	1.33	0.56	0.07

D. Discussion

The returned solution by nonlinear solvers in most cases is a local optimum or possibly no feasible solution, which means a global optimum is normally not guaranteed or practical. The results of this study suggest that although only a local solution was found, which proved to be optimal and, therefore, likely a global solution as well. In reality, for many planning applications, a technique with an achievable good local optimum provides more value and suffices as it is not absolutely necessary that the best local optimum found is definitely the global optimum [11]. It should be noted that in a distribution system without many meshed loops, such as that shown in Fig. 1, it is possible that a single local optimum exists. It has been demonstrated that for all cases studied, the CHP application cannot be employed at every node and the water network cannot support the complete recovery of exhaust heat. However, for the gas network, it can provide enough gas for the DG penetration and the CHP application functions well. For actual deployment, other concerns, such as investment, income, and location availability, should also be considered to size and locate the DG units [30].

VII. CONCLUSION

The developed generalized approach based on an integrated system model is novel as this is the first instance of allocating CHP-based DG based on the coupling of generalized urban energy distribution networks. The overall objective of maximizing the DG electrical output and recovered thermal output is different from cost minimization problems as more congestion may appear not only in the electrical network but also in the gas network or water network. The CHP-based DG units are placed at every node rather than being fixed onto some nodes. This approach is new and is being first employed in the DG capacity allocation based on urban energy networks.

APPENDIX

Electrical System Model							
AC power sources		Node ID					
ExGens		95	149	300	450		
Max/min real/reactive power		±60 MW		±50 MVAR			
System voltage level and total load			4.16 kV				
Real power		3.49 MW		Reactive power		1.92 MVAR	
Natural Gas System Model							
Gas main		Black lines		Grey lines			
Inner diameter (inches)		5.314		1.885			
Ambient temperature (°F)		70		System pressure (psig)		<60	
People per household		3.2		Average usage time (h)		3	
Average household gas usage (mcf)		19.52		Total system load (cm h)		880.33	
Water System Model							
Reservoir		Elevation (m)		5	Node ID		118
Water head (m) at node 250				50			
Nodal water head (m)			Max	50	Min	42.2	
People per household			3.2	Average usage time (h)			3
Hazen-Williams roughness coefficient				130			
Energy loss coefficient (globe type PRV)				5			
Pump operation coefficient				1.154×10^{-5}			
CHP-Based DG Model							
Microturbine C65		Electrical capacity (kW)			65		
		Thermal capacity (kW)			116.5		
DG power/fuel equation coefficient			A^{Co}		B^{Co}		C^{Co}
			-0.0203		11.483		94.13
Single unit water flow rate (m³/h)			Max	13.6	Min	5.68	
Sending water temperature (°F)			Max	220	Min	60	
Returning water temperature (°F)			Max	185	Min	85	
Heat exchanger efficiency			0.9				
Water heat capacity (J/(g* K)			4.18	Water density (kg/m³)		1000	

REFERENCES

- [1] Y. He, R. Sharma, and X. Zhang, "Microgrid operator's capacity and storage investment strategies under environmental regulations," in *Proc. IEEE PES Conf. Innovative Smart Grid Technol.*, Washington DC, USA, 2012, pp. 1–8.
- [2] P. Chiradeja and R. Ramakumar, "An approach to quantify the technical benefits of distributed generation," *IEEE Trans. Energy Convers.*, vol. 19, no. 4, pp. 764–773, Dec. 2004.
- [3] X. Zhang, G. Karady, and Y. Guan, "Design methods investigation for residential microgrid infrastructure," *Eur. Trans. Elect. Power*, vol. 21, no. 8, pp. 2125–2141, 2011.
- [4] *Methods for Calculating Efficiency*, [Online]. Available: <http://www.epa.gov/chp/basic/methods.html>.
- [5] X. Zhang, R. Sharma, and Y. He, "Optimal energy management of a rural microgrid system using multi-objective optimization," in *Proc. IEEE PES Innovative Smart Grid Technol. Conf.*, Washington DC, USA, 2012, pp. 1–8.
- [6] R. U. Ayres, H. Turton, and T. Casten, "Energy efficiency, sustainability and economic growth," *Energy*, vol. 32, no. 5, pp. 634–648, May 2007.
- [7] W. Graus and E. Worrell, "Trend in efficiency and capacity of fossil power generation in the EU," *Energy Policy*, vol. 37, no. 6, pp. 2147–2160, Jun. 2009.
- [8] M. Chaudry, J. Ekanayake, and N. Jenkins, "Optimum control strategy for a CHP unit," *Int. J. Distrib. Energy Resour.*, vol. 4, no. 4, pp. 265–280, 2008.
- [9] X. Zhang, G. Karady, K. Piratla, and S. Ariaratnam, "Nexus between distributed generation and urban water infrastructure," in *Proc. IEEE Power Eng. Soc. General Meeting*, pp. 1–8, 2012.
- [10] C. J. Dent, L. F. Ochoa, and G. P. Harrison, "Network distributed generation capacity analysis using opf with voltage step constraints," *IEEE Trans. Power Syst.*, vol. 25, no. 1, pp. 296–304, Feb. 2010.
- [11] C. J. Dent, L. F. Ochoa, G. P. Harrison, and J. W. Bialek, "Efficient secure ac opf for network generation capacity assessment," *IEEE Trans. Power Syst.*, vol. 25, no. 1, pp. 575–583, Feb. 2010.
- [12] G. Celli, E. Ghiani, S. Mocci, and F. Pilo, "A multiobjective evolutionary algorithm for the sizing and siting of distributed generation," *IEEE Trans. Power Syst.*, vol. 20, no. 2, pp. 750–757, May 2005.
- [13] W. El-Khattam, Y. Hegazy, and M. Salama, "An integrated distributed generation optimization model for distribution system planning," *IEEE Trans. Power Syst.*, vol. 20, no. 2, pp. 1158–1165, May 2005.
- [14] A. Piccolo and P. Siano, "Evaluating the impact of network investment deferral on distributed generation expansion," *IEEE Trans. Power Syst.*, vol. 24, no. 3, pp. 1559–1567, Aug. 2009.
- [15] W. El-Khattam, K. Bhattacharya, Y. Hegazy, and M. Salama, "Optimal investment planning for distributed generation in a competitive electricity market," *IEEE Trans. Power Syst.*, vol. 19, no. 3, pp. 1674–1684, Aug. 2004.
- [16] M. Geidl and G. Andersson, "Optimal power flow of multiple energy carriers," *IEEE Trans. Power Syst.*, vol. 22, no. 1, pp. 145–155, Feb. 2007.
- [17] M. Arnold, R. Negenborn, G. Andersson, and B. De Schutter, "Model-based predictive control applied to multi-carrier energy systems," in *Proc. IEEE Power Eng. Soc. General Meeting*, Calgary, AB, Canada, pp. 1–8, 2009.
- [18] C. Unsuhay-Vila, J. Marangon-Lima, A. de Souza, I. Perez-Arriaga, and P. Balestrassi, "A model to long-term, multiarea, multistage, and integrated expansion planning of electricity and natural gas systems," *IEEE Trans. Power Syst.*, vol. 25, no. 2, pp. 1154–1168, May 2010.
- [19] A. Martinez-Mares and C. Fuerte-Esquivel, "A unified gas and power flow analysis in natural gas and electricity coupled networks," *IEEE Trans. Power Syst.*, vol. 27, no. 4, pp. 2156–2166, Nov. 2012.
- [20] C. Unsuhay, J. Lima, and A. de Souza, "Modeling the integrated natural gas and electricity optimal power flow," in *Proc. IEEE Power Eng. Soc. General Meeting*, Tampa, FL, USA, pp. 1–7, 2007.
- [21] B. Awad, M. Chaudry, J. Wu, and N. Jenkins, "Integrated optimal power flow for electric power and heat in a microgrid," in *Proc. 20th Int. Conf. Exhibition on Electricity Distribution*, Prague, Czech Republic, 2009, pp. 1–4.
- [22] W. H. Kersting, *Radial distribution test feeders*, [Online]. Available: <http://ewh.ieee.org/soc/pes/dsacom/testfeeders/index.html>.
- [23] B. B. Bean, *PE, GASWORKS 9.0*, [Online]. Available: <http://www.b3pe.com/>.
- [24] *Residential natural gas consumption in Arizona*, [Online]. Available: <http://apps1.eere.energy.gov/states/residential.cfm/state=AZ>.
- [25] L. Carroll and R. Weston Hudkins, *Advanced Pipeline Design*, [Online]. Available: <http://www.ou.edu/class/che-design/a-design/projects-2009/Pipeline%20Design.pdf>.
- [26] L. W. Mays, *Hydraulic Design Handbook*, New York, NY, USA: McGraw-Hill, 1999.
- [27] *C65 & C65 ICHP MicroTurbine Natural Gas*, [Online]. Available: http://www.capstoneturbine.com/_docs/datasheets/C65%20&%20C65-ICHP%20NatGas_331035E_lowres.pdf.
- [28] Technical Reference, *Application Guide – Model C65 Integrated CHP (ICHP)*, CA, USA, Capstone Turbine Corporation, 2010.
- [29] J. Bisschop and M. Roelofs, *AIMMS – The Language Reference*, Haarlem, The Netherlands: Paragon Decision Technology, 2006.
- [30] G. Karady and X. Zhang, "Sustainability and resilience of electric energy supply in urban environment," in *Proc. IEEE PES Power Syst. Conf. Exposition*, Phoenix, AZ, USA, 2011, pp. 1–3.

Xianjun Zhang (S'10) received the B.S. degree in electrical engineering from North China Electric Power University, Beijing, China, in 2008, and the M.S. and Ph.D. degrees in electrical engineering from Arizona State University, in 2010 and 2013, respectively.

His research interests include distributed generation, design, planning, and energy management of energy microgrids, and electric power market analysis and design. He has worked as a full-time intern at NEC Laboratories America, Inc., and California ISO in the summer of 2011 and 2012, respectively. He currently works as a Market Design Engineer II at Midcontinent ISO.

George G. Karady (SM'70–F'78–LF'98) received the B.S.E.E. and Doctor of Engineering degrees in electrical engineering from Technical University of Budapest.

He is Power System Chair Professor at Arizona State University, Tempe, AZ, USA, where he teaches electrical power and performs research in power electronics, high-voltage techniques, and electric power systems. Previously, he was with EBASCO Services, where he served as Chief Consulting Electrical Engineer, Manager of Electrical Systems, and Chief Engineer of Computer Technology. He also served as Electrical Task Supervisor for the Tokomak Fusion Test reactor project at Princeton. Before that, he worked for the Hydro Quebec Institute of Research as a Program Manager. He started his career at the Technology University of Budapest, where he progressed from Post Doctoral Student to Deputy Department Head.

Dr. Karady is a registered professional engineer in New York. He is the author of more than 200 technical papers. He is active in IEEE; he was chairman of Chapter/Membership's Award Committee, Education committee's Award Subcommittee, and WG on nonceramic insulators, etc. He also served in the U.S. National Committee of CIGRE as Vice President and Secretary Treasurer.

Samuel T. Ariaratnam photograph and biography not available at the time of publication.